

Invisible to Visible: Alternative Human Interaction with Sensory-Enhancing Wearables

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Abstract

This paper explores the transformative potential of integrating technology with human sensory and emotional experiences through the "Outward Bound" idea. By utilizing responsive wearable devices, specifically the Sensitive Wing and its iterative developments[10], the project aims to externalize internal sensory experiences, bridging the gap between personal and external empathy. The fusion of traditional practices like Nüshu and ruffs with modern technology underscores the importance of preserving cultural heritage while embracing technological advancements.

This study highlights the complex interplay between gender, culture, and technology, visualizing the otherwise invisible tensions. Ultimately, these innovative approaches invite us to continuously and creatively reimagine our relationships with each other and the world, promising new perspectives on human interaction through technology.

CCS Concepts

• Applied computing; • Human-centered computing → Interaction devices;

Keywords

wearable technology, sensory design, design research, embodied interaction, affective computing, self-empowerment, auto-ethnography, autobiographical design

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1 Introduction

The development of tangible technologies signifies a seismic shift in our interaction with the world. These innovations are not merely tools but are reshaping human relationships, offering unparalleled connectivity while highlighting the complexities of this interconnectedness. Inspired by these advances, I proposed the Outward

Bound idea in 2023 to explore internal sensory experiences. By enhancing personal and external empathy, this concept aims to bridge emotional and experiential divides, fostering a renewed sense of connection with our surroundings.

Due to the perceived unreliability of human senses, an increasing amount of data is obtained through validated scientific instruments, with sensory data from the human body commonly being quantified [12]. Steven Shapin's groundbreaking description of science portrays it as a network of material, social, and literary technologies, where "each component depends on and integrates with other technologies." [13][14]. Traditionally, we consider instruments as external tools aiding in the derivation of objective scientific data. However, sensory-enhancing wearable technologies suggest that human physical capabilities can be extended through technology, integrating instruments into the human body.

Instead of relying solely on external instruments to generate scientific data and compress unnecessary data streams, these functions can be partially replaced by enhanced human capabilities. This requires developing new observational and interpretative skills. While some data may be lost, more data can be spontaneously sensed[11][7].

In essence, the integration of technology and design with human sensory and emotional experiences is an ongoing exploration of what it means to be human in a technologically advanced world. By redefining the boundaries between the self and the external environment, these innovative approaches invite us to reimagine our relationships with each other and the world, promising a future where technology enhances, rather than diminishes, our human experience.

2 Outward Bound

"Outward Bound" can be interpreted as the externalization of the senses, which involves transforming internal sensory experiences into tangible expressions accessible to the external world. Through creative expression, design, and technology, this process aims to make one's emotions, feelings, and thoughts more readily perceivable by others. This opens avenues for dialogue and interaction, breaking the invisible constraints imposed by social masking and self-monitoring [15]. In the existing frameworks of social interaction, individuals often hide their inner experiences, striving to conform to collective norms.

The "Outward Bound" design enriches interactions between individuals by transforming hidden internal sensory experiences into externally visible expressions. This visible impact is achieved through the use of sensory-enhancing wearable devices, allowing internal emotional states to be presented through visual signals,

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color changes, or other forms, enhancing transparency and face-to-face interaction between people. This expression does not need to be highly intuitive because the primary goal is to encourage deeper interpretation and engagement, rather than merely providing clear and immediate recognition of emotional states.

The potential of "Outward Bound" necessitates an examination of the internet's impact. The rise of social networking has shifted the locus of conversation and communication from the physical to the virtual realm. In this online world, individuals feel compelled to quantify themselves to align with the demands of online existence, yet engaging in meaningful, multidimensional debate through anonymous online communication remains a challenge[17]. American computer scientist and early virtual reality promoter, Jaron Lanier, has critiqued that users of social networks degrade themselves to access these services[8]. Furthermore, the rapid spread of COVID-19 since 2019 has exacerbated the decline of face-to-face communication.

I use the concept of "Outward Bound" to elucidate a specific approach to enhancing human senses through wearable devices. The primary goal is to provide innovative ways to explore and express internal sensory experiences, fostering empathy and connection with the external world. More importantly, it aims to reshape the relationship between humans and their surroundings, creating new forms of interaction and understanding.

3 Related work

To understand the practical applications and implications of "Outward Bound," we can examine the work of various designers and researchers who explore the intersection of technology, sensory enhancement, and human experience.

Iga Węglińska, a Polish multidisciplinary designer, investigates the relationship between the human body and clothing[18]. Her Emotional Clothing project reveals psycho-physical changes in the wearer's body through biofeedback. Similarly, Leah Heiss's Ether Beat creates a space for empathy at a distance through innovative clothing, exploring unconventional methods of perception and communication[6].

Innovative practices in Outward Bound often draw on the 'extended mind' theory by Andy Clark and David Chalmers [3]. This theory challenges the view of the mind as confined to the brain, proposing it as a distributed system that includes both biological and technological components. According to Clark and Chalmers, tools and technologies become extensions of our minds.

In this context, Aposema is a speculative design project that addresses potential future challenges in human empathy[1]. This facial wearable device, positioned at the intersection of art, science fiction, and product design, speculates that as empathy diminishes, society might use masks to externalize moods and maintain social continuity.

Examining these innovative practices and theoretical frameworks provides insights into how Outward Bound can reshape our understanding of human interaction and empathy through the externalization of sensory experiences.

This project includes an EEG feedback device presented as an interactive wearable, exploring non-verbal visualization of physiological data. Using an iterative self-ethnographic method, I deconstruct my own discourse and practice, making the invisible visible and maintaining a continuous tension.



Figure 1: Wear Sensitive Wings, Nüshu script represents 'invisible gender'

4 Design

In this section, I report on two main aspects of the design process: how the practice has iterated based on previously published foundations[10], and their production and implementation.

4.1 Making the Invisible Visible

Practice-based research to promote reflection in action[2]. In the practice of the Sensitive Wing, I went through different iterations. These iterations did not enhance the clarity of this dialogue capability but rather established a layer of implicit communication. This is because connectivity does not always signify direct communication; sometimes, it's about ambiguous exchanges. From the initial simple data visualization to the deconstruction of Chinese characters, the most recent version incorporates Nüshu. This process continually reveals the power structures and assumptions behind it. Complex emotions are also conveyed and perceived through this ambiguity.

Since the 16th century, ruffs have been standalone entities tied around the neck. High-ranking European nobles were obsessed with larger and more elaborate designs as they represented wealth, social status, and prestige. Former Queen Elizabeth I further legislated the size of collars worn in her kingdom and even banned the use of blue collars[4]. The ruff, as a fabric continuing the logic of hierarchy and status through formal variations, reveals that the flow of institutions and power is constructed by material entanglements.

I have drawn from this form and combined it with another form of constrained communication, Nüshu. Nüshu is the world's only existing script system developed by and for women, serving as a unique carrier of female spirituality and consciousness. Historically, due to various social and cultural constraints, women in China had limited opportunities for formal education. Consequently, Nüshu

became a vital tool for literary creation, documenting daily life, communicating emotions, and passing on knowledge. Women used sticks as pens and potash as ink, with knowledge being transmitted from mother to daughter, elder to younger, and teacher to apprentice[5].

However, according to local customs, Nüshu scripts kept by women were often burned after their passing, leading to the scarcity of surviving texts. The widespread adoption of Chinese characters also caused newer generations of women to abandon this script. A promotional board in the Nüshu Museum in Jiangyong profoundly moved me. It listed many women's names, whose education levels (in terms of Chinese literacy) were officially defined as illiterate in household documents, yet their proficiency in Nüshu was marked as expert. My attempts to find literature on Nüshu in both English and Chinese yielded very few results[16].

The invisible gender, script, and brainwaves have been transformed into visible oscillations. The oscillation is tumultuous, its meaning ambiguous.

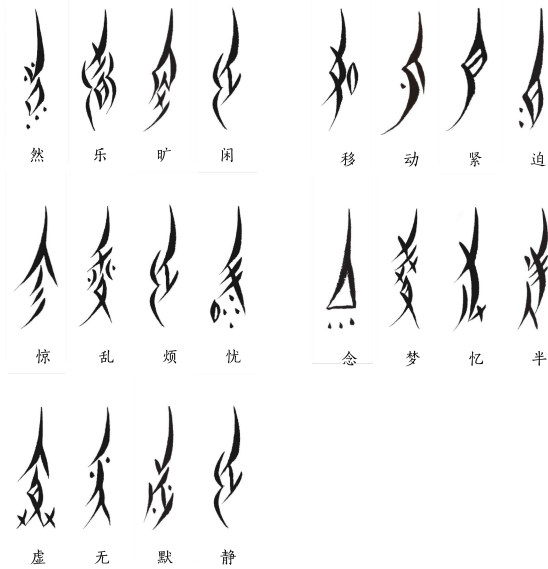


Figure 2: 20 adopted Nüshu characters and their Hanzi characters

4.2 System Design

The main structure is made from a piece of cotton fabric measuring 700 cm by 7 cm, meticulously sewn by hand after being treated with cyanotype. It is built around the Neurosky ThinkGear ASIC Module (TGAM) and supplemented with components such as a servo system and a Bluetooth transceiver. Additionally, the setup includes one piece of cyanotype fabric measuring 200 cm by 40 cm and two pieces measuring 200 cm by 70 cm to create a temporary communication space.

This method relies on waveforms that can be recognized by the EEG sensors. From these waveforms, five notable frequency bands are extracted: Gamma, Beta, Alpha, Theta, and Delta [9]. In this

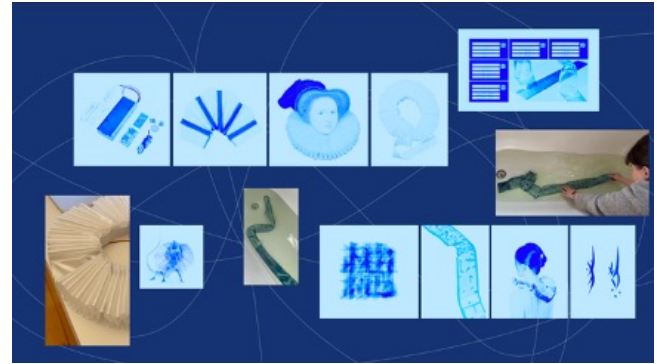


Figure 3: The non-linear process of creation and iteration

process, a total of 20 different Nüshu characters were converted to represent the potential meanings of different waveforms. Each waveform then generates 3 or 6 hexadecimal data points for comparison. To determine the rotation angle along a specific axis, the hexadecimal data is first converted to decimal format and then the individual values are summed. The final number is normalized through byte count to derive the interactive components of the rotation.

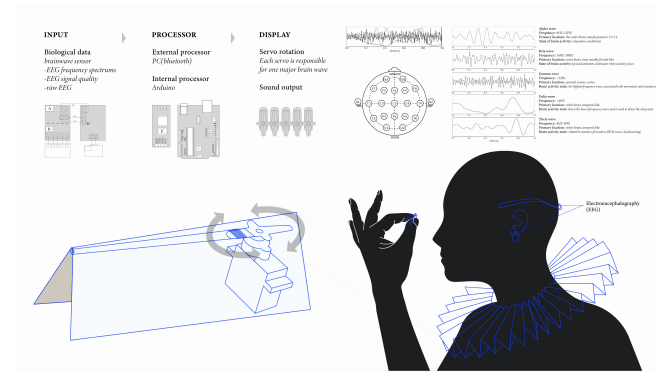


Figure 4: The technical mapping of Sensitive Wing

5 Conclusion

This practice-based research demonstrates the transformative potential of integrating technology with human sensory and emotional experiences. The "Outward Bound" concept, utilizing responsive wearable devices, enables the externalization of internal sensory experiences. This approach redefines our interactions by bridging the gap between personal and external empathy, enhancing our connections with each other and our environment.

Combining traditional practices like Nüshu with modern technology highlights the importance of preserving cultural heritage while embracing technological advancements. This fusion reveals the complex interplay between gender, culture, and technology, providing new perspectives on how we experience and interpret the world.

Future work will continue to employ self-ethnographic methods to reflect on and refine current practices. Enhancing the integration of EEG detection devices will improve wearer comfort and coherence. Additionally, I will explore further possibilities for wearable devices within the "Outward Bound" context.

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